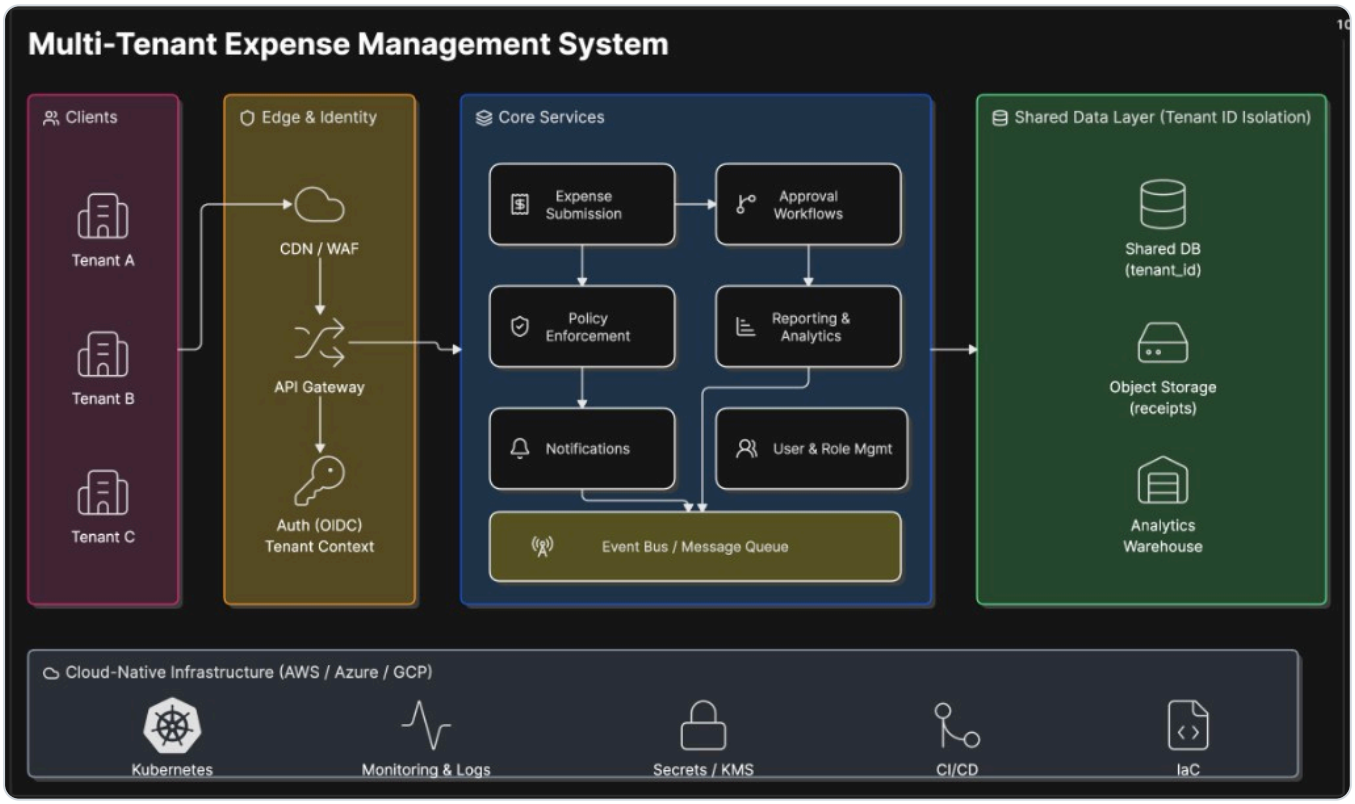
EMS — Technical Documentation (API Reference & UI Walkthrough)

This document describes the implemented Expense Management System: every API endpoint with request/response samples, with **UI screenshots shown inline next to the flow they belong to**.

- **Base URL:** `http://localhost:4000`
- **Format:** JSON over HTTP
- **Source design:** `level-2-design.md` · scope: `level-1.5-scope.md`

Architecture

Target / production architecture — multi-tenant clients behind an edge + API gateway, stateless core services on a shared data layer with `tenant_id` isolation, on cloud-native infra:



The as-built prototype implements the same core capabilities as a **modular monolith** (single Express app + PostgreSQL + optional Redis); the gateway, event bus, object storage, analytics warehouse and infra layers are the documented production path.

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1. Conventions

Authentication

- Obtain a token via `POST /auth/login` or `POST /auth/signup`.
- Send it on every protected request: `Authorization: Bearer <accessToken>`.
- The token encodes `{ id, org_id, role, email }`. The server derives `org_id` from the token — clients never pass it. This is the tenant-isolation guardrail.

Roles & permissions

Permission	employee	manager	finance	admin
expense:create / read own	✅	✅	✅	✅
expense:read reportees	—	✅	—	✅
expense:read all	—	—	✅	✅
expense:approve	—	✅	✅	✅
policy:manage / budget:manage	—	—	✅	✅
user:manage	—	—	—	✅
analytics:view	—	✅	✅	✅

Headers

Header	When	Purpose
`Authorization: Bearer <token>`	all protected routes	authentication
`Content-Type: application/json`	requests with a body	parsing
`Idempotency-Key: <unique>`	approve/reject (optional)	prevents double-apply on retry

Error envelope

All errors share one shape:

```
{ "error": { "code": "FORBIDDEN", "message": "You do not have permission...", "details": null } }
```

HTTP	code	Meaning
400	BAD_REQUEST	Malformed/invalid payload
401	UNAUTHORIZED	Missing/invalid token
403	FORBIDDEN	Authenticated but not allowed
404	NOT_FOUND	Resource not in your tenant
409	CONFLICT	Invalid state transition / duplicate
422	UNPROCESSABLE	Valid shape but business rule failed (budget, currency, no policy)

Notes on data types

- Postgres `NUMERIC` columns are returned as **strings** (e.g. `"3000.00"`), preserving precision. Clients should parse as needed.
- Pagination: list endpoints accept `?limit=` (max 100) and `?offset=`.

2. Auth & Onboarding

`POST /auth/signup` — create a new organization (tenant onboarding)

Atomically creates an organization, its first **admin** user, a default approval policy and an org budget, then returns tokens (auto-login). Public.

Request

```
{
  "orgName": "Initech",
  "baseCurrency": "INR",
  "adminName": "Bill Lumbergh",
  "email": "admin@initech.test",
  "password": "password123"
}
```

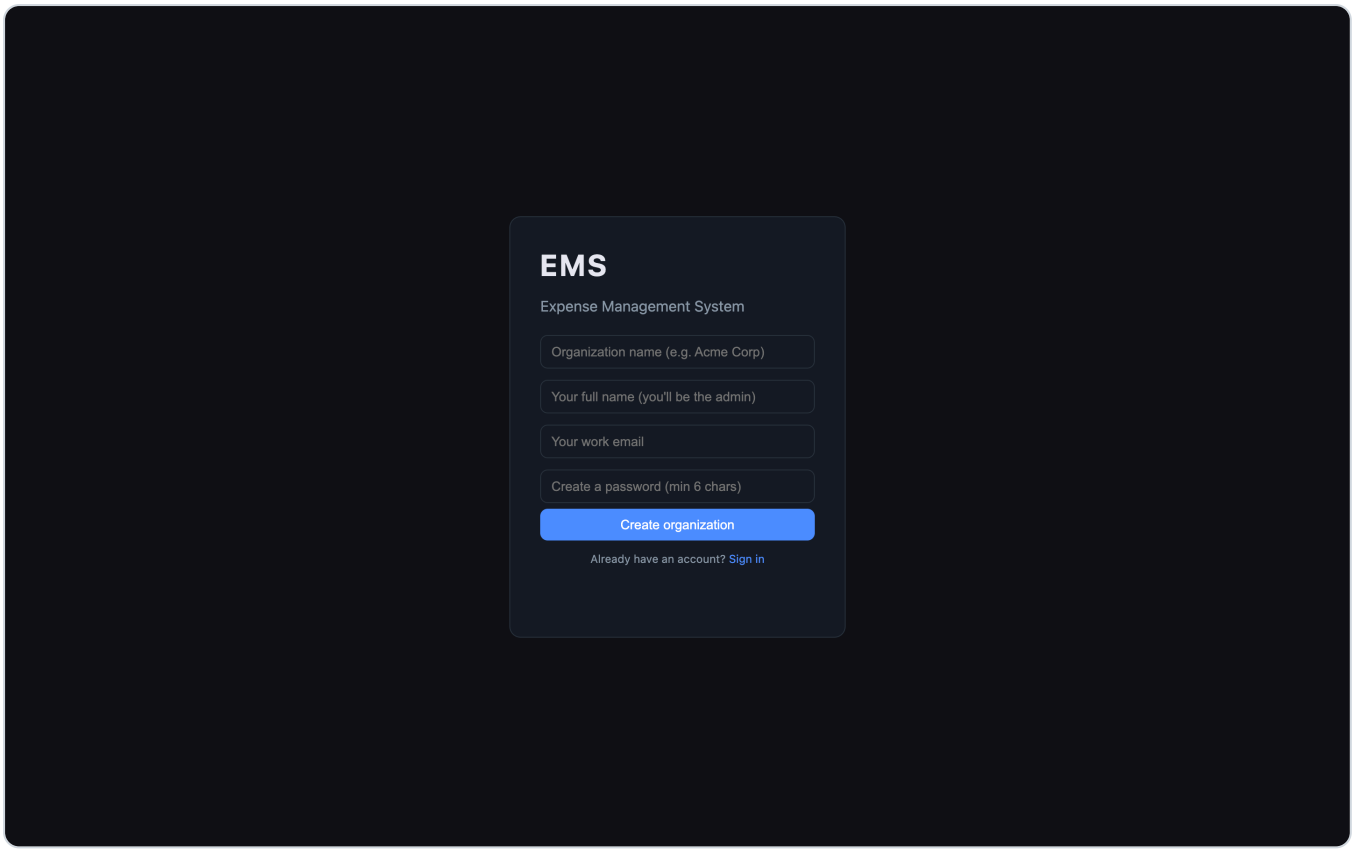
Field	Type	Required	Notes
orgName	string	yes	organization name
baseCurrency	string(3)	no	defaults `INR`
adminName	string	yes	first admin's name
email	string	yes	unique across all tenants
password	string	yes	min 6 chars

Response `201`

```
{
  "accessToken": "eyJhbGci...",
  "refreshToken": "eyJhbGci...",
  "user": { "id": "uuid", "org_id": "uuid", "name": "Bill Lumbergh", "email": "admin@initech.test",
  "organization": { "id": "uuid", "name": "Initech", "base_currency": "INR" }
}
```

Errors: `400` invalid payload · `409` email already exists.

UI — Sign up (new organization): "Create an organization" provisions a new tenant + its first admin in one step, with a default policy so it's usable immediately.



`POST /auth/login`

Request

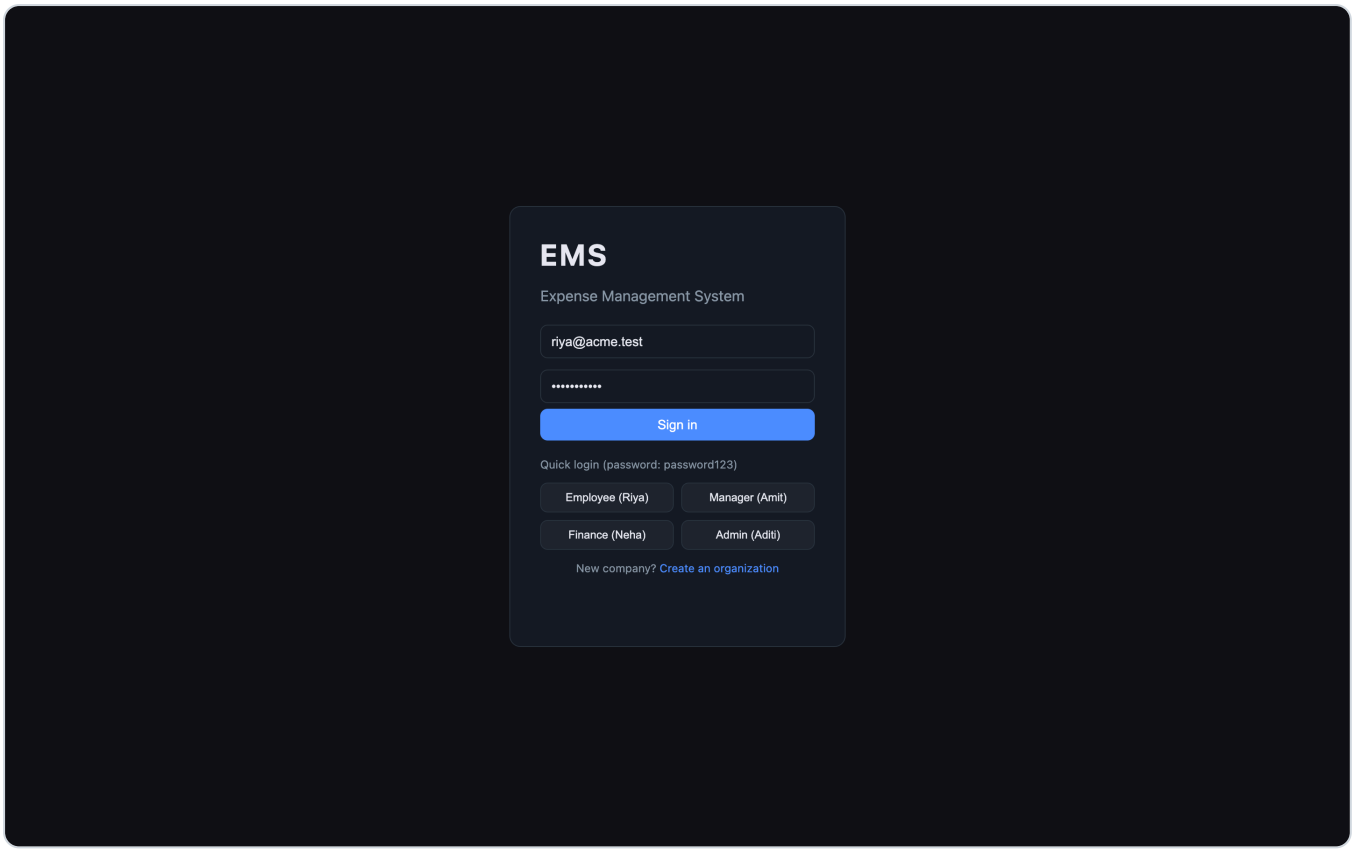
```
{ "email": "riya@acme.test", "password": "password123" }
```

Response `200`

```
{
  "accessToken": "eyJhbGci...",
  "refreshToken": "eyJhbGci...",
  "user": {
    "id": "uuid", "org_id": "uuid", "name": "Riya Sharma",
    "email": "riya@acme.test", "role": "employee",
    "manager_id": "uuid", "is_active": true, "created_at": "2026-..."
  }
}
```

Errors: `401` invalid email or password.

UI — Login: sign in with email/password, or one-click **Quick login** for the seeded roles.



`POST /auth/refresh` — rotate refresh token

Validates an opaque refresh token, **revokes it (single-use rotation)** and issues a fresh access + refresh pair. Replaying a revoked token is treated as reuse and revokes all of the user's sessions.

Request

```
{ "refreshToken": "<opaque token from login/signup>" }
```

Response `200`

```
{ "accessToken": "eyJhbGci...", "refreshToken": "<new opaque token>", "user": { "id": "uuid", "...":
```

Errors: `401` missing / invalid / expired / reused token.

`POST /auth/logout`

Revokes a refresh token. Idempotent. **Request:** `{ "refreshToken": "<token>" }` · **Response** `200` → `{ "revoked": true }`

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3. System

`GET /health`

Response `200` → `{ "status": "ok" }`

`GET /metrics`

Prometheus text exposition. Public.

```
ems_requests_total 128
ems_errors_total 0
ems_request_latency_ms{quantile="0.5"} 4
ems_request_latency_ms{quantile="0.95"} 21
ems_request_latency_ms{quantile="0.99"} 39
```

---

4. Expenses

`POST /expenses` — create a draft

Permission: `expense:create`.

Request

```
{ "type": "reimbursement", "category": "travel", "description": "Client visit", "amount": 22000, "cu
```

Field	Type	Required	Notes
type	enum	yes	`reimbursement` | `company_paid`
category	string	no	defaults `general`
description	string	no	
amount	number	yes	> 0, in `currency`
currency	string	no	defaults `INR`; supported: INR, USD, EUR, GBP, AED, SGD

> The `category` values shown in the UI come from `GET /categories` (admin-managed per org). An expense **cannot be created until the org has an active approval policy** — otherwise there is nothing to route it through.

Response `201`

```
{
  "id": "uuid", "org_id": "uuid", "requester_id": "uuid",
  "type": "reimbursement", "category": "travel", "description": "Client visit",
  "amount": "22000.00", "currency": "INR", "base_amount": "22000.00", "fx_rate": "1.000000",
  "status": "draft", "policy_snapshot_json": null, "current_level": 0,
  "sla_due_at": null, "created_at": "2026-...", "updated_at": "2026-..."
}
```

Errors: `400` invalid · `422` unsupported currency · `422` no active policy configured for the org.

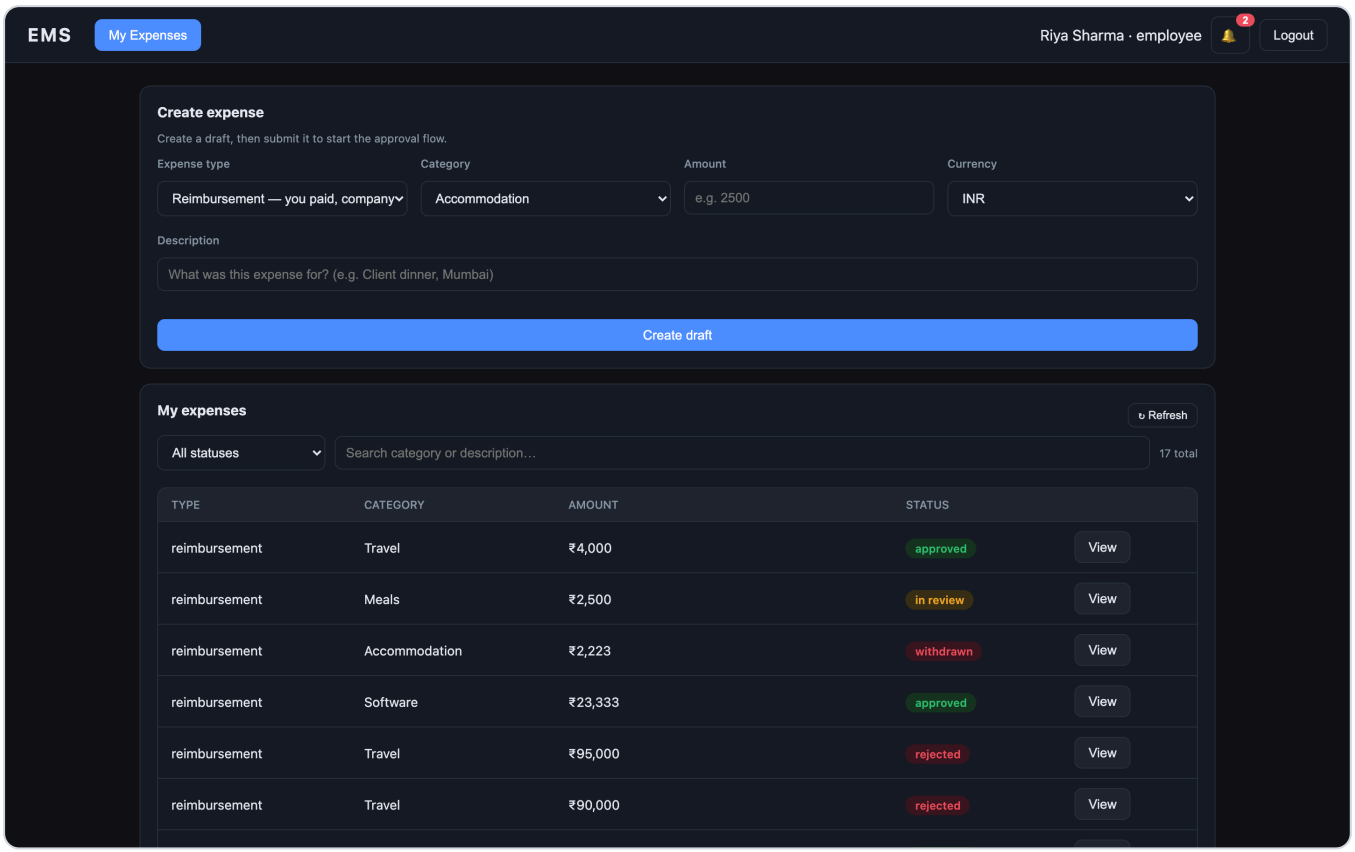
`GET /expenses` — list

Query: `scope=mine|reportees|all` (default `mine`), `status`, `type`, `limit`, `offset`.

- `reportees` requires `expense:read:reportees`; `all` requires `expense:read:all`.

Response `200` → array of expense objects (newest first). **Errors:** `403` if scope exceeds your permission.

UI — Employee "My Expenses": create reimbursement or company-paid requests and track them with status pills. Foreign-currency amounts show the base (₹) conversion inline.



`GET /expenses/:id` — get one (with approval chain)

Response `200`

```
{
  "id": "uuid", "status": "in_review", "current_level": 1, "amount": "22000.00",
  "base_amount": "22000.00", "type": "reimbursement", "category": "travel",
  "steps": [
    { "id": "uuid", "level": 1, "required_role": "manager", "approver_id": "uuid", "status": "approv
    { "id": "uuid", "level": 2, "required_role": "finance", "approver_id": "uuid", "status": "pendir
  ]
}
```

Errors: `404` not found / not in your tenant · `403` not allowed to view.

`PATCH /expenses/:id` — edit

Only the requester, only while `draft` / `submitted` / `in_review`. If the amount crosses a policy threshold while `in_review`, the approval chain is **rebuilt**. **Request** (any subset): `{ "amount": 20000, "currency": "INR", "category": "...", "description": "..." }` **Response** `200` updated expense. **Errors:** `403`, `409`, `422`.

`DELETE /expenses/:id`

Requester only, drafts only. **Response** `200` → `{ "deleted": true }`. **Errors:** `403`, `409`.

`POST /expenses/:id/submit`

Moves `draft → in_review`, snapshots the policy, builds the `ApprovalStep` chain, and notifies **only the first approver**. Approvals are **sequential**: the next level's approver is notified only after the current level approves; a rejection ends the chain. Before building the chain the engine verifies an **eligible approver exists for every level** (excluding the requester) and fails fast otherwise. **Response** `200` → expense with `status:"in_review"`, `current_level:1`. **Errors:** `403` not requester · `409` not a draft · `422` budget exceeded / no active policy / **no eligible approver for a required level**.

Budget-exceeded body:

```
{ "error": { "code": "UNPROCESSABLE", "message": "Budget exceeded for monthly (limit 2000000, alread
  "details": { "ok": false, "period": "monthly", "limit": 2000000, "spent": 1990000 } } }
```

`POST /expenses/:id/approve`

Permission: `expense:approve`. Must be the assigned approver for the current level; cannot approve your own (approver resolution already routes an expense to a *different* eligible user, so it never lands in the requester's own queue). Approving the last level marks the expense `approved`; otherwise it advances `current_level` and notifies the next approver. Honors `Idempotency-Key`. **Request:** `{ "reason": "looks good" }` (optional). **Response** `200` → expense; becomes `approved` after the last level, else stays `in_review` with `current_level` advanced. **Errors:** `403` wrong approver / self-approval · `404` · `409` not awaiting approval.

`POST /expenses/:id/reject`

Permission: `expense:approve`. **Reason is mandatory.** Terminates the chain. **Request:** `{ "reason": "not a business expense" }` **Response** `200` → expense `status:"rejected"`. **Errors:** `400` no reason · `403` · `404` · `409`.

`POST /expenses/:id/withdraw`

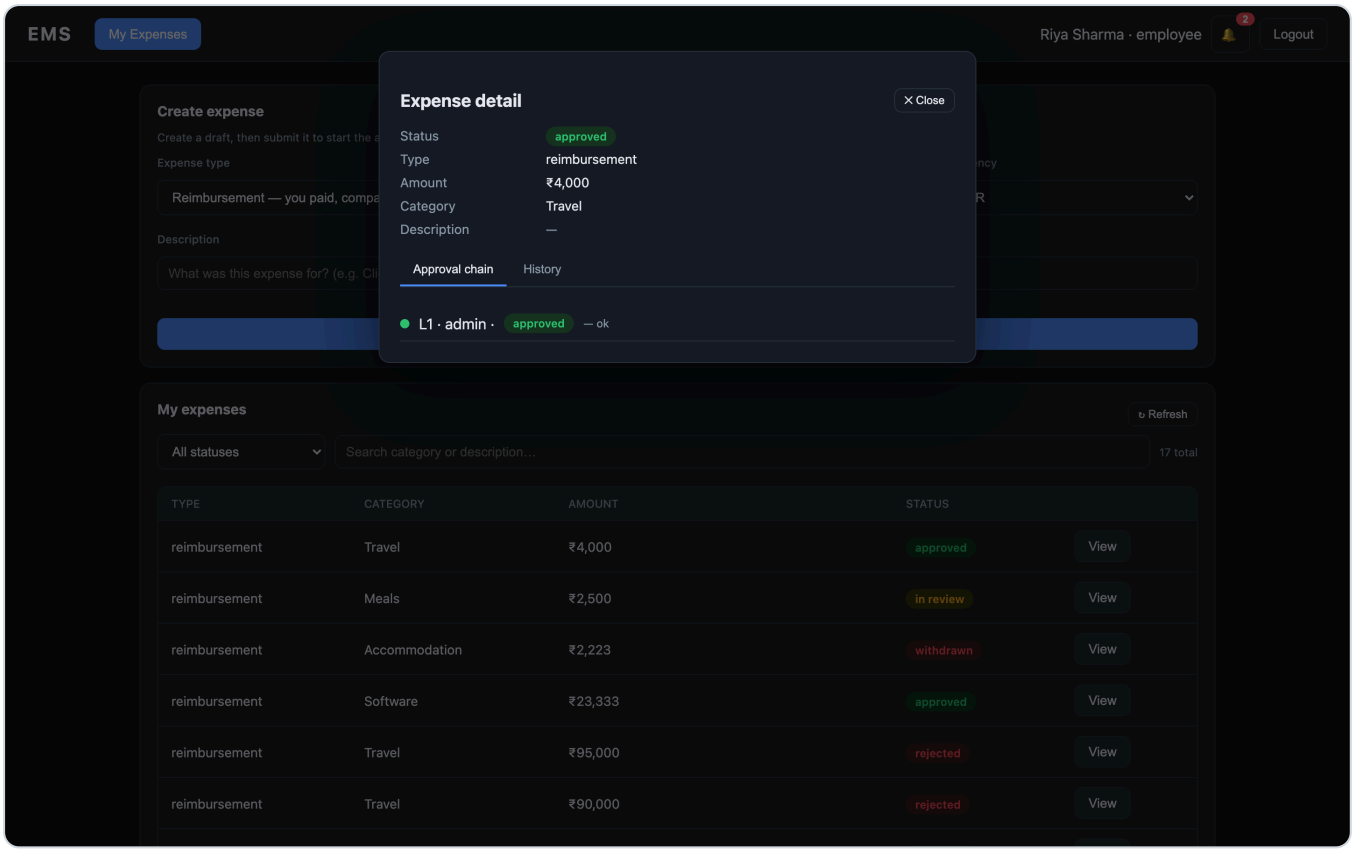
Requester cancels (`draft` / `submitted` / `in_review`). **Response** `200` → `status:"withdrawn"`.

`GET /expenses/:id/history` — audit trail

Response `200`

```
[
  { "id":"uuid","actor_id":"uuid","action":"expense.created","entity_type":"expense","entity_id":"uu
  { "action":"expense.submitted", "after_json":{"status":"in_review","levels":["manager","finance"]}
  { "action":"expense.step_approved", "reason":"ok", "created_at":"2026-..." },
  { "action":"expense.approved", "created_at":"2026-..." }
]
```

UI — Expense detail (chain + audit trail): each expense shows its computed multi-level **approval chain** (level, role, status) and the immutable **history (audit trail)**. Requester actions (Submit/Withdraw) appear contextually.



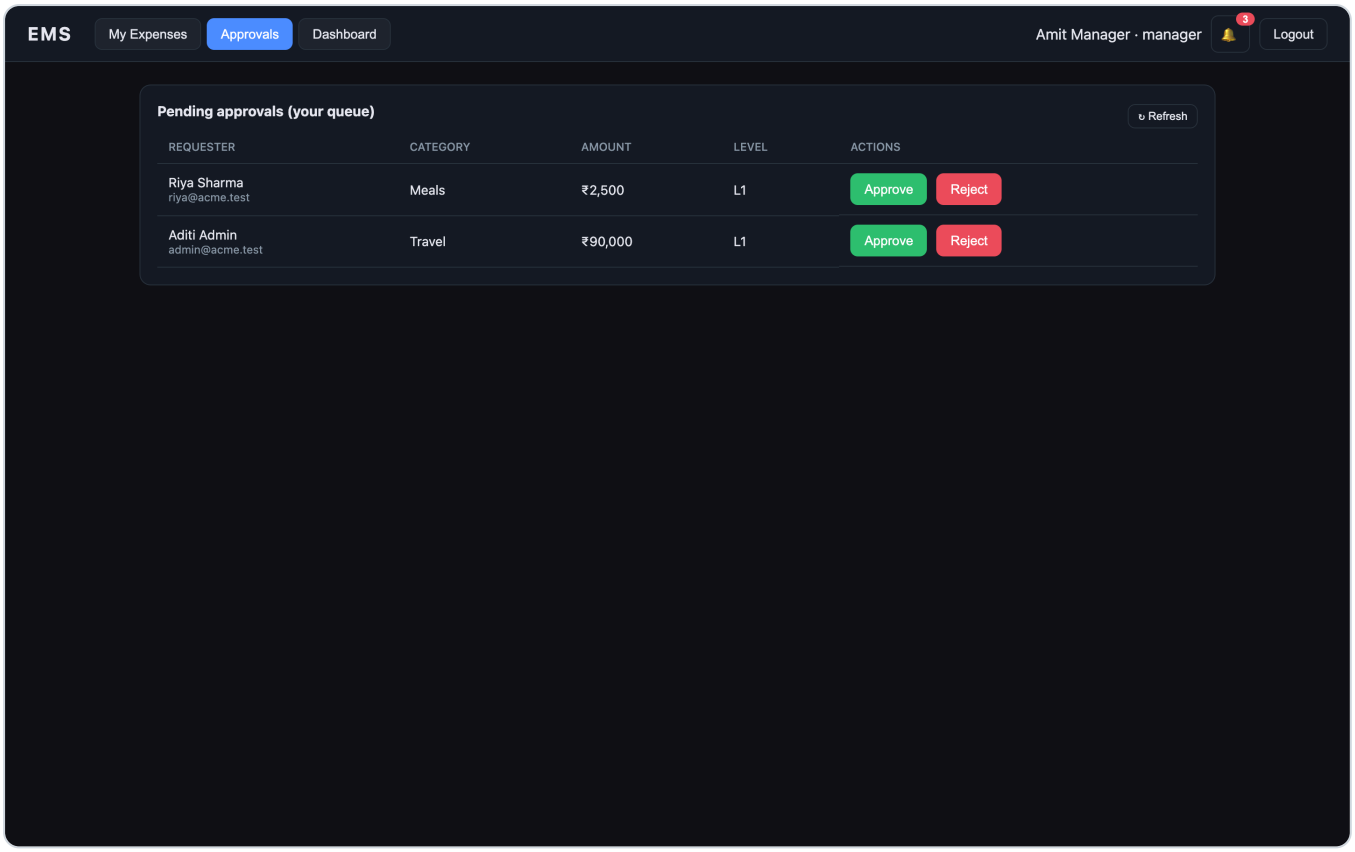
5. Approvals

`GET /approvals/pending` — my approval queue

Permission: `expense:approve`. Returns steps assigned to you that are pending at the expense's current level. **Response `200`**


```
[
  { "id":"step-uuid","expense_id":"uuid","level":1,"required_role":"manager","status":"pending",
    "amount":"4500.00","base_amount":"4500.00","currency":"INR","type":"reimbursement",
    "category":"meals","description":"Client dinner","requester_id":"uuid",
    "requester_name":"Riya Sharma","requester_email":"riya@acme.test","expense_status":"in_review" }
]
```


UI — Manager approvals queue: a manager sees only the steps assigned to them at the current level, and can Approve or Reject (reason prompted) directly.



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6. Attachments

`POST /attachments/presign` — mock S3 presigned URL

Request: `{ "filename": "bill.png", "contentType": "image/png" }` **Response** `200`

```
{
  "key": "ems-attachments/<org>/<uuid>-bill.png",
  "uploadUrl": "https://mock-s3.local/...?X-Amz-Signature=mock&expires=900",
  "method": "PUT", "expiresInSeconds": 900,
  "note": "MOCK presigned URL (no real S3 in prototype)"
}
```

Errors: `400` missing filename/contentType.

---

7. Policies

`GET /policies`

Permission: `analytics:view`. **Response** `200` → array:

```
[{ "id":"uuid","name":"Default approval policy","tolerance_percent":"10.00","active":true,"version":
  "rules_json":{ "currency":"INR","rules":[
    {"min":0,"max":5000,"levels":["manager"]},
```

```
    {"min":5001,"max":50000,"levels":["manager","finance"]},
    {"min":50001,"max":null,"levels":["manager","finance","admin"]}]}}]
```

> **Exactly one policy is active per org.** The approval engine always uses the single active policy, so the UI never shows several "active" rows competing to route an expense.

`POST /policies` — create

Permission: `policy:manage`. Creating a policy as `active` (the default) **deactivates every other policy in the org in the same transaction**. Policy names must be unique within the org. **Request**

```
{ "name": "Travel policy", "tolerancePercent": 10,
  "rulesJson": { "currency":"INR", "rules":[ {"min":0,"max":10000,"levels":["manager"]} ] } }
```

Response `201` policy object. **Errors:** `400` invalid · `409` duplicate policy name · `422` a rule's `max < min`.

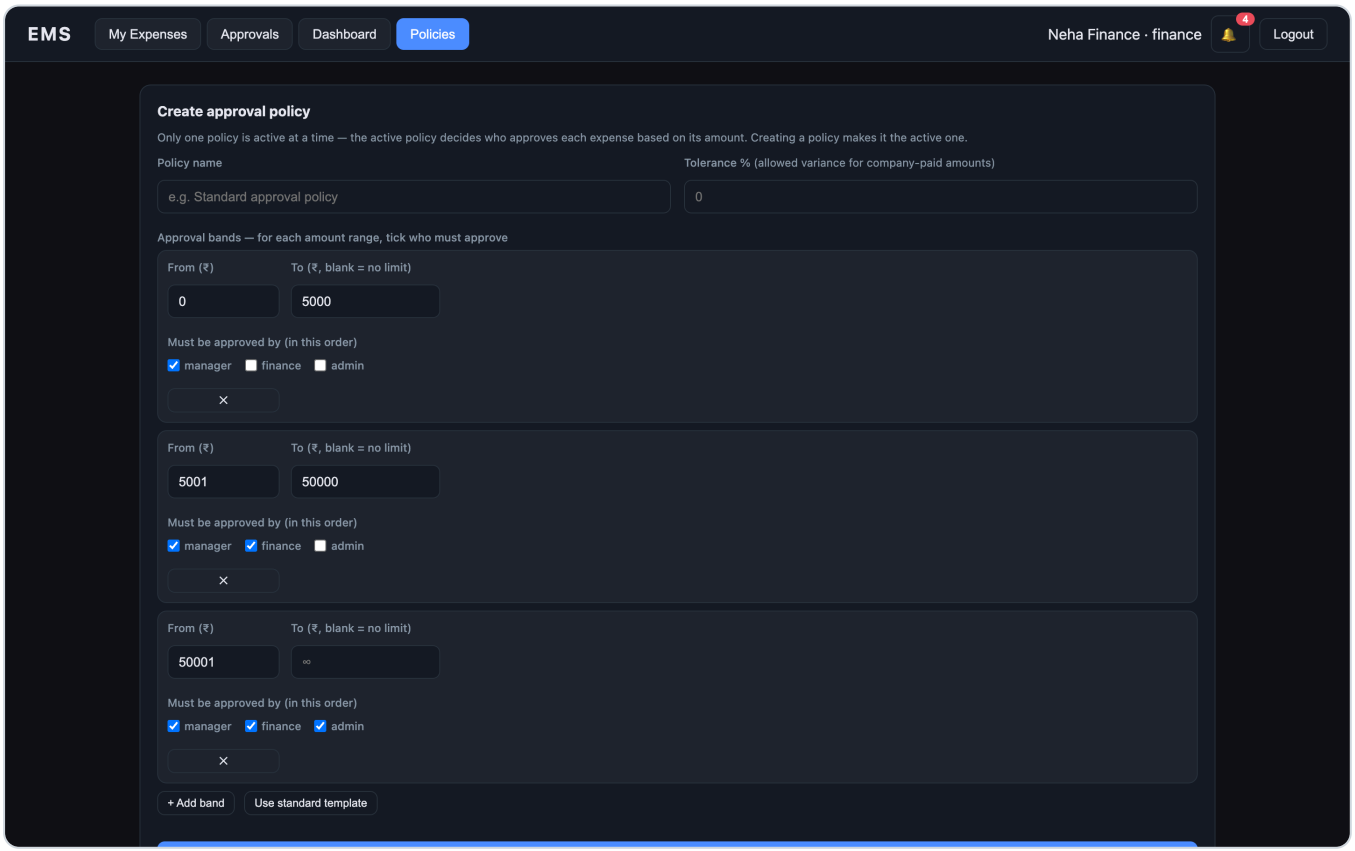
`PATCH /policies/:id` — update (bumps `version`)

Any subset of `name`, `tolerancePercent`, `active`, `rulesJson`. Setting `active:true` activates this policy and **deactivates all others** atomically. **Errors:** `404` if missing · `409` duplicate name.

`DELETE /policies/:id`

Response `200` → `{ "deleted": true }`. **Errors:** `404` if missing · `400` the **active policy can't be deleted** (activate another one first).

UI — Policies: a **visual rule builder** (amount band → ordered approver chain, no JSON to hand-write) plus the tolerance percentage. The list shows which policy is active, lets you activate/deactivate with one click, and delete inactive ones. Admins also manage the org's **expense categories** here (the list that feeds the expense-form dropdown).



7b. Categories

Per-org expense categories that populate the category dropdown on the expense form. Seeded with sensible defaults on signup; admins can add/remove more.

`GET /categories`

Permission: any authenticated user. **Response** `200`

```
[{ "id":"uuid","org_id":"uuid","name":"travel","active":true,"created_at":"2026-..." }]
```

`POST /categories` — add a category

Permission: `policy:manage`. **Request:** `{ "name": "software" }` **Response** `201` category object. **Errors:** `400` invalid · `409` duplicate name.

`DELETE /categories/:id` — soft-deactivate

Permission: `policy:manage`. Marks the category inactive (history on existing expenses is preserved). **Response** `200` → `{ "deleted": true }`. `404` if missing.

---

8. Budgets

`GET /budgets` (perm `analytics:view`) → array of budgets.

`POST /budgets` (perm `budget:manage`)

Request: `{ "userId": "uuid|null", "scope": "user|org", "period": "daily|monthly", "limitAmount": 50000, "currency": "INR" }` **Response** `201` budget object. **Errors:** `400` invalid.

`GET /budgets/utilization` — current user's usage

Response `200` → `{ "monthly": { "limit": 2000000, "spent": 25200, "pct": 1 } }`

---

9. Users

`GET /users` (perm `user:manage`) → array (password hashes stripped).

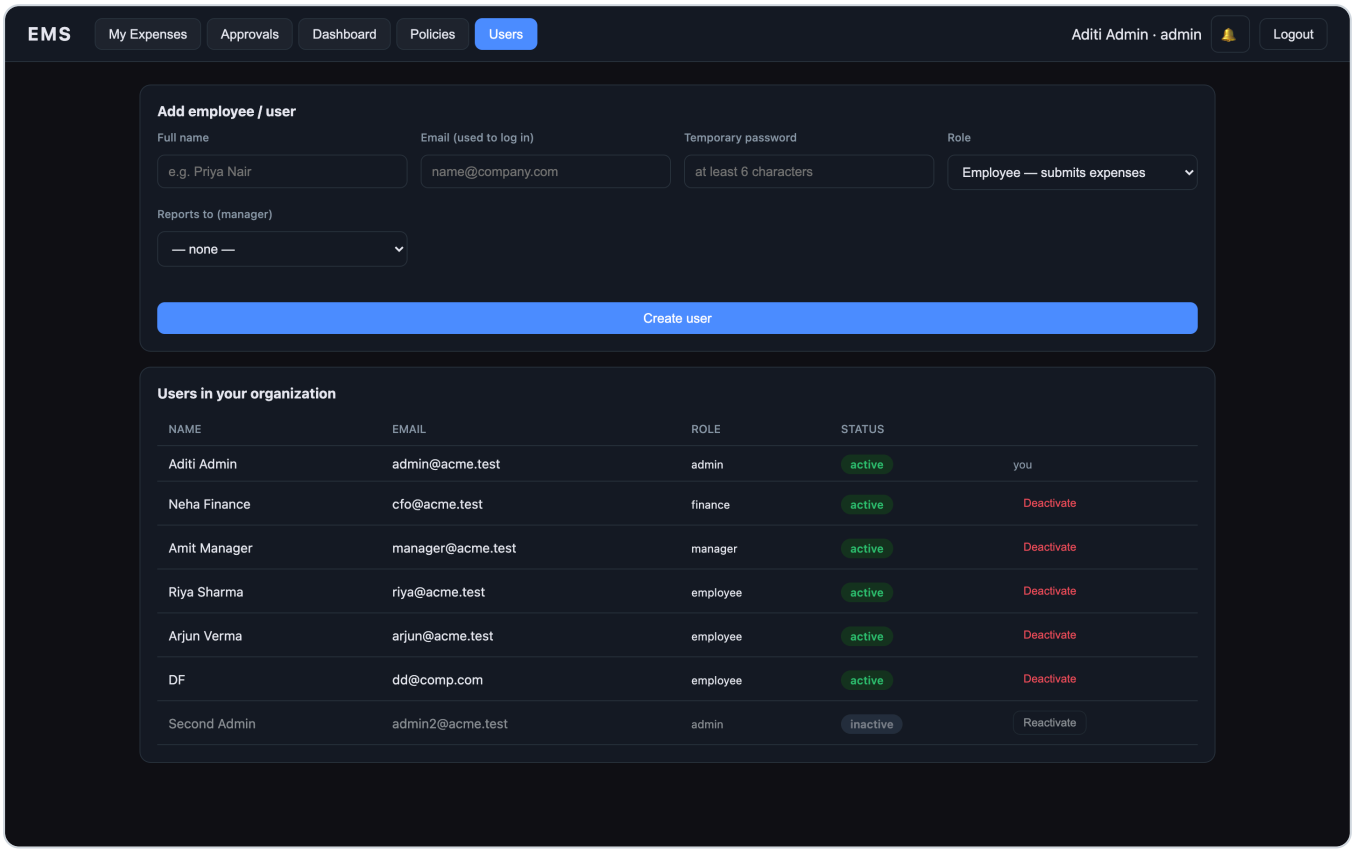
`POST /users` (perm `user:manage`)

Request: `{ "name":"New", "email":"new@acme.test", "password":"secret1", "role":"employee", "managerId":"uuid|null" }` **Response** `201` user (no hash). **Errors:** `400` invalid · `409` duplicate email.

`PATCH /users/:id` (perm `user:manage`)

Any subset of `name`, `role`, `managerId`, `isActive`. Setting `isActive:false` **deactivates** the user — they can no longer log in and are skipped by approver resolution; `isActive:true` reactivates them. **Guards:** you cannot deactivate **your own account**, nor the **last active admin** in the org (→ `400` / `409`). `404` if missing.

UI — Admin Users: admins add employees/managers/finance to their org (with optional manager assignment), view the org roster with active/inactive status, and deactivate or reactivate users from the table.



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10. Notifications

`GET /notifications` → my notifications (newest 100).


```
[{ "id":"uuid","type":"approval_requested","payload_json":{"expenseId":"uuid","level":1},"read":fals
```


`POST /notifications/:id/read` → `{ "read": true }`. `404` if not yours.

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11. Analytics / Observability

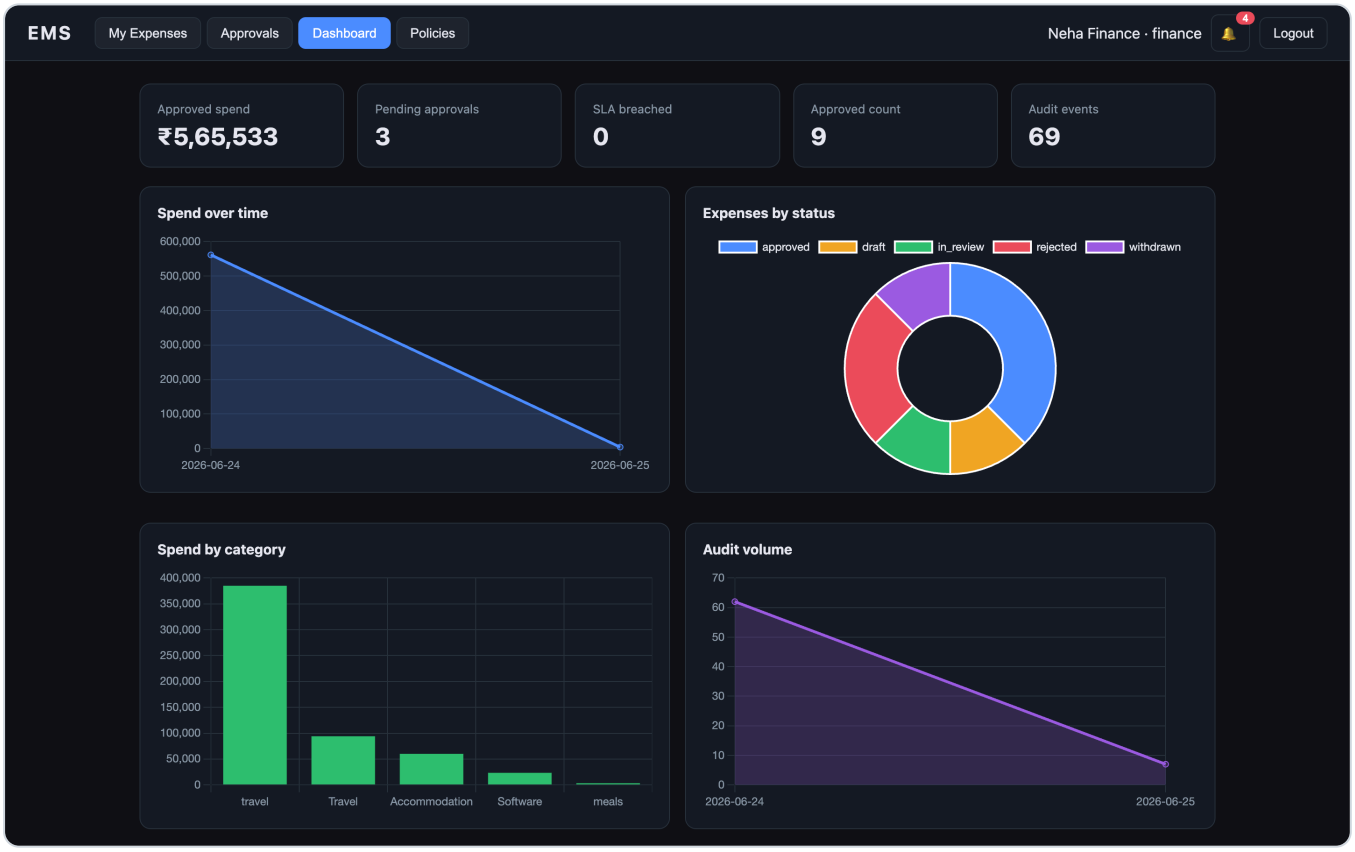
All require `analytics:view`. Summary is Redis/memory-cached (short TTL).

Endpoint	Response
`GET /analytics/summary`	`{ approvedSpend, approvedCount, pendingApprovals, slaBreached, auditEvents, byStatus:{...} }`
`GET /analytics/spend`	`[{ "day":"2026-06-24", "total": 25200 }]`
`GET /analytics/by-status`	`[{ "status":"approved", "count": 2 }]`
`GET /analytics/by-category`	`[{ "category":"travel", "total": 22000 }]`
`GET /analytics/audit-volume`	`[{ "day":"2026-06-24", "count": 16 }]`

Example `GET /analytics/summary`:

```
{ "approvedSpend": 25200, "approvedCount": 2, "pendingApprovals": 2, "slaBreached": 0,
  "auditEvents": 16, "byStatus": { "approved": 2, "draft": 1, "in_review": 2, "rejected": 1 } }
```

UI — Finance/Admin observability dashboard: stat cards (approved spend, pending, SLA breached, audit events) plus charts — spend over time, expenses by status, spend by category, audit volume. Data is aggregated from Postgres and cached.



12. Security, isolation & concurrency

Authentication. Access tokens are short-lived JWTs (`{ id, org_id, role, email }`). Refresh tokens are **opaque random strings stored only as SHA-256 hashes** in `refresh_tokens` . `/auth/refresh` does **single-use rotation** (old

token revoked, new one issued, chained via `replaced_by`); replaying a revoked token is treated as theft and revokes the user's whole session family. `/auth/logout` revokes a token. *(Code: `src/auth/refreshToken.ts`.)*

Tenant isolation. `org_id` is read from the verified token (never the client) and applied to every query — the core isolation guardrail. This is app-layer isolation; one missing predicate would leak across tenants. The production hardening is **Postgres Row-Level Security**:

```
ALTER TABLE expense_requests ENABLE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON expense_requests
  USING (org_id = current_setting('app.org_id')::uuid);
-- set per transaction:  SET LOCAL app.org_id = '<org from token>';
```

It's documented rather than enabled here because RLS interacts with the connection pool and the migration/seed paths; the app-layer guard plus this plan is the deliberate tradeoff.

Concurrency (no double-apply). Every state transition (submit/edit/approve/reject/withdraw) opens a transaction and takes an exclusive row lock via `SELECT … FOR UPDATE` (`getByIdForUpdate`). Concurrent approvals on the same expense are serialized by Postgres: the loser blocks, re-reads the committed state, and is rejected by the status/level guards. Idempotency keys are orthogonal — they dedupe identical retries, the row lock handles distinct concurrent requests.

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13. End-to-end flow (recap)

```
signup/login → create expense (draft) → submit
   → engine reads policy, snapshots it, builds ApprovalStep chain by amount
   → notify approver(s)
manager approve → (advance level) → finance approve → ... → APPROVED
                      |
                  reject (reason) → REJECTED
every transition → immutable audit_log entry (in the same DB transaction)
finance/admin → dashboard reads aggregates for live observability
```